FORM HS-03 · REV. 5 POST-STASIS REFRESHER · MODULE 3 OF 9 POST AT EVERY MUSTER POINT

DUNEWRIGHT COLONY WORKS
Settlement Site 4 · Construction Division · Safety & Health

HEAT KILLS.

THIS WORLD IS HOTTER THAN ANYTHING YOUR BODY REMEMBERS. READ EVERY LINE BEFORE YOUR FIRST SHIFT.

JUST WOKE UP?

YOUR HEAT TOLERANCE DID NOT SURVIVE STASIS.

However many seasons you worked in heat before, **you are a new worker now.** Your body rebuilds heat tolerance from zero over 7–14 days. Most heat deaths happen in a worker's **first few days** on a hot site. Before your first shift, Revival Med Bay weighs you in, checks your hydration and issues your RAMP band.

Day 1 is your first shift with any heat exposure after Revival Med Bay clears you.

20%	40%	60%	80%	100%
DAY 1	DAY 2	DAY 3	DAY 4	DAY 5
½ TIME	½ TIME	½ TIME	½ TIME	FULL TABLE

Each bar is the most of your shift you may spend in the heat. **On Days 1–4, also work half the table's minutes** and rest for the rest of the hour: 40/20 becomes 20/40, 30/30 becomes 15/45, 10/50 becomes 5/55, and "No limit" becomes 30/30 (site rule). **Day 1: light work only**, no heavy lifting, digging or hauling (site rule). Spend the rest of a ramp shift in cool-room training or on light duty out of the heat. Your crew lead logs your ramp day on the roster.

- **Days 6–14: acclimatizing.** Full table. At WBGT 85 or higher, your crew lead checks you every 15 min.
- **When the ramp applies:** whenever the heat level is Caution or higher, or the WBGT is 78 or higher.
- **Reset to Day 1** (RAMP band back on) after a fever, vomiting or diarrhea, any heat illness, or 7 or more days away from heat work. It applies to every worker, however long ago they were revived (site rule). After an illness absence, Revival Med Bay clears you again first.
- **Medicines and health.** Tell Revival Med Bay at clearance, or the FAS later, about any medicine you take (water pills, allergy pills, blood-pressure or heart pills, some mood medicines, stimulants) and about a past heat stroke or any heart, kidney or blood-sugar problem. Your ramp may change.

RAMP band: black-and-yellow chevron band reading "RAMP", with a reflective strip, worn on the front and back of your helmet. On at Day 1. It comes off only after your crew lead logs Day 5 complete. See one on a helmet? Watch out for that worker.

SITE HEAT LEVELS

POSTED AT MUSTER AND EVERY HOUR
BY YOUR CREW LEAD · CH 16 BROADCASTS CHANGES

Each level keeps every rule from the levels below it. At every level: buddy pairs, anyone with heat symptoms is pulled, the work/rest table applies whenever the WBGT is 78 or higher, and every active deck has a stocked cooling station with an ice trough.

70-79°	**NORMAL** Normal ops	Water on your belt. Heavy suits and hard labor can still drop you. **Watch anyone wearing a RAMP band.**
80-89°	**CAUTION** Heat plan active	Cool drinking water and shade within reach of every crew. 8 oz every 15–20 min. Ramp rules apply. **No lone work** (site rule).
90-102°	**EXTREME CAUTION** Buddies in sight	At least a **15-min break in shade or a cooling station every 2 hours**. Buddies stay within sight of each other. Crew lead walks the line.
103-124°	**DANGER** Restricted work	Heavy and very heavy work moves to the night shift. Work/rest by the table below. **No working ice trough on a deck: stop work on that deck** until it's fixed (site rule).
BELOW THIS LINE: 125° AND HOTTER, AS AIR TEMPERATURE. SITE RULES. OLD EARTH HEAT CHARTS END AT THE NWS EXTREME DANGER BAND.		
125-139°	**EXTREME** Essential work only · site rule	Essential work is work the Sector Superintendent approves for that shift: life safety or stopping damage. **Cooling vest mandatory.** Buddies in sight. No more than 30 min outside, then at least 30 min in a cooled shelter. Swap or recharge your vest and drink before you go back out. The air is hotter than your skin, so wind heats you and fans on dry skin don't help.
140-159°	**EXTREME HEAT ENTRY** Heat Entry Permit PTW-HE · site rule	No one goes out without a **Heat Entry Permit (PTW-HE)**, signed by the area general foreman and countersigned by a Safety officer (REF-PTW). A PTW-HE is not a hot-work permit. Cooled suit or supplied cool air. The permit sets each entry time, never more than 15 minutes, then at least 30 minutes in a cooled space. Every entry is logged on the permit board. A standby rescuer in full gear waits at the entry point. A medic checks pulse and temperature after every entry: a pulse over 120 one minute after stopping, or a temperature over 101.3°, ends that worker's entries for the shift (ACGIH basis). **Only workers who have finished the ramp and Days 6–14, and are cleared for heat entry that shift by Revival Med Bay or the FAS, may enter (site rule).**
160-180°	**SITE CLOSED** Emergency only	All crews to shelter. Ch 16 calls the move when the forecast will reach 160° within the hour. Don't wait for the reading. Steel, rebar and deck plate **burn skin in about one second**. Entry only for rescue, in full reflective cooling gear with a timed return.

Which number? Levels up to 124° use the NWS heat index in shade. From 125° up they use air temperature in shade. Your crew lead posts the heat level and the WBGT. **Follow whichever of the two is more restrictive.** WBGT runs much lower than air temperature or heat index, so never read a WBGT number against these levels: a WBGT of 90° is near the top of the work/rest table, while a heat index of 90° is only Extreme caution.

HEAT STROKE

Anyone who collapses or isn't fully alert on a hot site is a heat-stroke case until a medic says otherwise. Confused, slurring, stumbling, seizing, hot to the touch. They may still be sweating. Don't wait for a thermometer: only a medic's rectal reading counts.

COOL FIRST. TRANSPORT SECOND.

Start cooling within minutes. Brought below about 102° within 30 minutes of collapse, nearly everyone survives (Korey Stringer Institute). The medic decides when to stop cooling.

1. **Ch 1: "Emergency, emergency, emergency. Heat casualty"**, then location (Sector-grid, structure, deck, landmark), number hurt, hazards and your call sign. Stay on Ch 1 until Hub Command or the medic releases you.
2. **Get them out of the heat** into the cooling station. Strip off suit, vest and helmet.
3. **Ice trough, water up to the neck.** Hold the head up and keep the airway clear. Stir the water.
4. No trough? **Douse them** with the coldest water you have, without stopping. Lay ice-water towels on them and swap each one for a freshly iced towel every 2–3 min. Ice at the neck, armpits and groin. Fan only while the skin is soaking wet.
5. **No drinks if they're confused.** Never leave them. Keep cooling until the medic takes over.

KNOW THE SIGNS

IN YOURSELF AND IN YOUR CREW

HEAT RASH

Clusters of red bumps on the neck, chest or groin.

`DO:` keep the skin dry, cool off, use powder. No creams.

HEAT CRAMPS

Painful spasms in the legs, arms or belly during heavy sweating.

`DO:` rest in shade, drink water with electrolyte mix. Go to the FAS if cramps last over an hour, or right away if you have a heart condition or are on a low-salt diet.

FAINTING

Dizzy or light-headed after standing still or getting up fast.

`DO:` sit or lie down in the cool, sip water, tell your crew lead. **Anyone who passes out, or isn't fully alert when they come to, is a heat-stroke case: Ch 1 and start cooling.**

HEAT EXHAUSTION

Heavy sweat, clammy skin, headache, nausea, dizziness, weakness, short temper, thirst, dark urine.

`DO:` pull them off the line into a cooling station. Cool the head, face and neck. Give sips of water. **Someone walks them to the Forward Aid Station (FAS) now, never alone.** Done for the day. Revival Med Bay clears them before their next shift. Confused or getting worse: it's heat stroke, call Ch 1.

RHABDO

Rhabdomyolysis, muscle breakdown: muscle pain and weakness, urine dark like tea or cola after hard work.

`DO:` stop work. Someone walks you to the FAS now. It can wreck your kidneys.

HEAT STROKE

Confusion, slurred speech, collapse, seizures, hot skin. They may still be sweating.

`DO:` follow the red box. Cool them on the spot.

Any heat illness resets the ramp to Day 1. Report every case, even cramps or a dizzy spell (see "Report it" below).

WORK / REST BY WBGT

MINUTES OF WORK / REST PER HOUR · WATER PER HOUR
OLD EARTH DATA: TOOL-HFC

WBGT (wet-bulb globe temperature) is the heat-stress number your crew lead measures where you work. It counts air temperature, humidity, sun and wind together. Find your WBGT row and your work type. Each cell shows **minutes of work / minutes of rest per hour**, then water per hour.

DCW TABLE: THE STRICTER OF TB MED 507 (2003 AND 2022) IN EVERY CELL

WBGT °F	LIGHT	MODERATE	HEAVY	VERY HEAVY
78–81.9	No limit · ½ qt	No limit · ¾ qt	40/20 · ¾ qt	20/40 · 1 qt
82–84.9	No limit · ½ qt	50/10 · ¾ qt	30/30 · 1 qt	15/45 · 1 qt
85–87.9	No limit · ¾ qt	40/20 · ¾ qt	30/30 · 1 qt	10/50 · 1 qt
88–89.9	No limit · ¾ qt	30/30 · ¾ qt	20/40 · 1 qt	10/50 · 1 qt
90–91.9	50/10 · 1 qt	20/40 · 1 qt	10/50 · 1 qt	10/50 · 1 qt
92–94.9 SITE RULE	50/10 · 1 qt	10/50 with cooling vest · 1 qt	**Not outside cooled spaces**	**Not outside cooled spaces**
95+ SITE RULE	**Essential work only, under Extreme-level rules: Sector Superintendent approval, cooling vest, buddies in sight, 30 min out then at least 30 min in a cooled shelter.**			

Follow the more restrictive of this table and your heat level. If your crew lead's local WBGT is higher than the Ch 16 value, the local reading wins. No working meter: use the Ch 16 WBGT for your grid, read the table one row hotter, and get a meter from the Quartermaster on Ch 15 (site rule).

Ramp: on Days 1–4, work half the minutes shown (see "Just woke up?"). Days 6–14 use the full table, with a crew lead check every 15 min at WBGT 85+.

Water: full sun adds about ¼ qt per hour. Never drink more than 1½ qt (48 oz) in an hour or 12 qt in a day.

Heavy or non-breathable gear (chemical suit, FR leathers, sealed coverall): add 10° to the WBGT for light work, or 20° for moderate, heavy or very heavy work (very heavy: site rule), before you read the table. Cooled suits: follow the PTW-HE.

Rest means sitting or standing still in a cooling station or shade, with gear loosened. **No limit** means no limit to work per hour, up to 4 hours straight. The times and water amounts hold for at least 4 hours of work.

The 92+ and 95+ rows are site rules. The Old Earth table's top row is 90 and above. At 92–94.9, moderate work needs a cooling vest. No vest, no work.

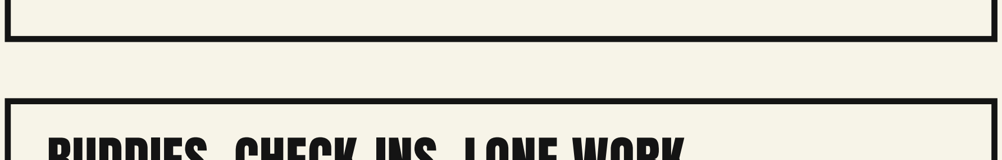
Light: inspection, survey, operating from an AC cab. **Moderate:** framing, tying rebar, carrying panels, walking on loose sand with a light load, welding (FR leathers add the clothing adjustment; never count them twice). **Heavy:** trenching, shoveling, concrete finishing, climbing. **Very heavy:** hauling or carrying heavy loads by hand on loose sand, two-person panel carries, fast digging.

WATER DISCIPLINE

8 OZ	48 OZ	2%
EVERY 15–20 MIN	MAX PER HOUR	WEIGHT LOST = PULL OFF

- Drink before you're thirsty. Thirst runs behind your losses.
- No more than 48 oz (1½ qt) an hour or 12 qt a day. More thins your blood salt (hyponatremia).
- Use the electrolyte drink mix, or eat your rations, with your water. **No salt tablets** unless a medic gives them to you.
- Weigh in and out at the **shift weigh station** in the cooling station. Measure against that day's starting weight (your revival weight is too low to use). Down 2% (3.6 lb for a 180-lb worker): pull off, rehydrate in the cooling station and check in at the FAS before you go back.
- Drink 16–24 oz for every pound lost, spread over the next few hours.
- Confused after drinking a lot? Don't give more water. Call Ch 1. The medic tells heat stroke apart from low blood salt.
- No alcohol within 24 hrs of a shift. No stim drinks on deck.

Drinking water and toilets: drink only from blue-labeled **DRINKING WATER** taps and coolers. Recycled water and coolant are never drinking water. Coolers stay shaded and insulated, with water at 50–60°F. Use your own bottle. Toilets are within a short walk of every deck, so never drink less to avoid a trip.



ON THE BUILD

- **Metal burns.** Sun-baked steel, rebar and deck plate pass 158°. Gloves on every time you touch metal.
- **Pours, lifts and heavy work go at night.** Concrete, heavy rigging and welding in leathers get scheduled for the cool hours.
- **Enclosed modules trap heat.** Unfinished habs and hull sections can run far hotter than outside. Your crew lead takes the WBGT inside before anyone goes in. One way in or poor airflow? Check it against Supplement A (HS-SA) first.
- **Cover up.** Loose, light-colored, full-coverage clothing, a brimmed hard-hat shade and a neck flap. Bare skin soaks up heat when the air is hotter than you.
- **Sun and UV.** Atmos posts the UV index on Ch 16. Skin burns in minutes. Cover up, put sunscreen on any bare skin, and wear tinted, UV-rated safety glasses or goggles. Burned skin sweats poorly.
- **Wet cloth works in dry air.** A soaked neck wrap cools you through evaporation. It stops working when it's humid.
- **Never sit on bare sand or deck plate.** Sit on a mat or a cooling-station bench.

BUDDIES, CHECK-INS, LONE WORK

- **Buddy pairs always.** Buddies watch each other all shift. From Extreme caution up, buddies stay within sight of each other.
- Your **crew lead** sees every crew member at least every 30 min, and checks in with the foreman every 30 min on the foreman talkgroup.
- **Missed check-in:** immediate call-back. No answer within 5 minutes: someone goes in person.
- **Lone work:** none at Caution or higher, at height, in confined spaces, or near wadis or trenches. Otherwise only with 15-minute check-ins and a logged location.

REPORT IT

Report every injury, illness, first aid case, near miss, damage and unsafe condition to your crew lead before the end of the shift (immediately for anything serious), on form HS-IR with your Colonist ID. That includes heat cramps and dizzy spells. Don't move anything except to rescue. Safety & Health investigates with the crew and shares the findings at muster. No retaliation.

CREW LEADS & FOREMEN

- Take the **WBGT where the crew works** (same deck, same sun or shade) at muster and every hour, and inside any enclosed module before entry. Post the heat level, the WBGT and the work/rest cycle.
- Track each worker's **ramp day** on the roster. Log Day 5 complete before a RAMP band comes off. Reset anyone who qualifies to Day 1. No one skips the ramp because the schedule slipped.
- Keep **water, shade and a cooling station** on the same deck as the work, within 2 minutes' walk, with no ladder climb in between (site rule).
- Check your deck's **ice trough** at muster and every 2 hours: full, ice floating, clean, and big enough for an adult lying down. Low or warm: call the Quartermaster on Ch 15. Clean it after every use.
- Brief the **heat casualty plan** at every muster: the nearest cooling station and ice trough, the Ch 1 call ("Emergency, emergency, emergency. Heat casualty"), and the route to the FAS. Ch 14 is for medics. The first call always goes on Ch 1.
- At Extreme, get the Sector Superintendent's approval before any essential work.
- Plan heavy tasks for the night shift. **Cut scope before you cut rest.**

STOP-WORK AUTHORITY

Anyone can stop unsafe work. No penalty, no questions. Stop it, make it safe, tell your crew lead (your foreman if there's no crew lead; any green hat if it isn't fixed). Work restarts only when the crew lead confirms the fix with the person who stopped it. Every stop is logged. Stop work when anyone shows heat symptoms, or when water, shade, a cooling station or the ice trough isn't there.

WORKER ACKNOWLEDGEMENT

I have read this poster, I know where my nearest cooling station and ice trough are, and I understand that I must follow the heat ramp.

NAME	COLONIST ID	CREW / SECTOR	DATE

Adapted from Old Earth standards: NIOSH Criteria for a Recommended Standard, Occupational Exposure to Heat and Hot Environments (2016-106); OSHA proposed Heat Injury and Illness Prevention rule (2024, not final); NOAA National Weather Service heat index; US Army TB MED 507 (2003, and April 2022 Table 3-2); ACGIH heat stress and strain TLV; CDC/NIOSH heat illness first aid; Korey Stringer Institute exertional heat stroke guidance.

HS-03 · Rev. 5 · Dunewright Colony Works · See also HS-01 · HS-09 · REF-PTW · TOOL-HFC · TOOL-WX · Replace when faded